The Malfatti circles in Lean 4, with independent certificate proofs of Feuerbach, Miquel and Brahmagupta

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Abstract

We formalise four classical theorems about circles in Lean 4 with Mathlib. Only one of them, the Malfatti circle construction, is to our knowledge formalised here for the first time; the other three have earlier formal proofs and are offered as independent formalisations. *Feuerbach's theorem* (number 29 on Wiedijk's list of 100 theorems): the nine-point circle of a triangle is internally tangent to the incircle and externally tangent to the three excircles, stated with Mathlib's own `Affine.Simplex.insphere`, `Affine.Simplex.exsphere`, `Affine.Simplex.ninePointCircle` and tangency predicates, together with Euler's formula $OI^2 = R(R - 2r)$, its excentral analogue and Euler's inequality $2r \le R$. *Miquel's theorem*: for points on the three (extended) sides of a triangle, none at a vertex, the three Miquel circles pass through a common point. *Brahmagupta's formula*: a convex cyclic quadrilateral with sides a, b, c, d has (shoelace) area $\sqrt{(s-a)(s-b)(s-c)(s-d)}$, the formal hypotheses also admitting some collapsed configurations for which the formula holds trivially, derived from Mathlib's Ptolemy theorem through the diagonal form of Bretschneider's formula, which we prove for every quadrilateral. *The Malfatti circles*: every triangle in the Euclidean plane has three circles of positive radius, each tangent to the two sides through one vertex and any two externally tangent, with Malfatti's radius formula $\rho_A = \frac{r}{2(s-a)}\big(s - r - (IB + IC - IA)\big)$; this is the construction, *not* Malfatti's conjecture that these circles maximise the total area, which Goldberg disproved in 1967 [6]. In each case the core is an explicit polynomial identity closed by `ring` or `linear_combination`. The four libraries use no `sorry` and depend only on Lean's three standard axioms; each passes an axiom gate, a tamper test and a further battery of statement-level checks. All four libraries were also rebuilt on a second machine of a different architecture (arm64/macOS). Version 2 adds two further layers: each of the four theorems restated with Mathlib-only types and an explicit non-degenerate witness, and a comparison with Lean statements written by a model from textbook statements alone. The comparison produced proofs that are new to these libraries (among them, that each Malfatti circle lies inside the triangle, and the angle form of Bretschneider's formula for convex quadrilaterals), found one writer statement false as written, and found the libraries narrower in two respects (Brahmagupta's shoelace area has not been formally identified with Lebesgue measure; uniqueness of the Malfatti circles is not proved). **Feuerbach's theorem is not new to Lean:** Weiyi Wang has a sorry-free standalone Lean repository (January 2026) and an open Mathlib pull request (#44236, created 26 September 2026). Our Feuerbach development is an independent second formalisation, and we claim no

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priority for it. **Miquel's theorem is not new to formal mathematics either:** it was formalised in Isabelle/HOL in the Archive of Formal Proofs entry of Freitas Ramos, Barros Hulak and de Queiroz (August 2026), and earlier in Coq's HighSchoolGeometry library. Our Miquel statement differs in assuming only the non-degeneracy conditions that make the three circles exist. **Brahmagupta's formula is not new to Lean:** a complete, sorry-free Lean 4 proof is in the ShadowBench repository (`epfl-lara/icml-26-lean-challenges`, commit of 12 June 2026). What our Brahmagupta development adds is limited to statements we did not find there: the inequality $K^2 \le (s-a)(s-b)(s-c)(s-d)$ and the equivalence of equality with Ptolemy's equality, both for arbitrary four points. For the Malfatti circle construction we found no earlier formalisation; our searches are not exhaustive, and three of the four searches missed existing work. The new contributions are therefore the Malfatti formalisation, a second Lean proof of Feuerbach's theorem, a Miquel statement with fewer hypotheses than the earlier ones, and the verification methodology with its limits.

1 Introduction

Circles are where elementary geometry becomes awkward to formalise. A statement such as "the three circles pass through a common point" presupposes that the circles exist and are unique; "the area of a cyclic quadrilateral" presupposes an order of the vertices and a notion of area; "the nine-point circle touches the incircle" must decide what happens for an equilateral triangle, where the two circles coincide. A formal statement has to settle each of these questions, and the settlement is part of the result.

This paper continues an earlier one on Desargues, Monge, Simson, Napoleon and Lami [10] and uses the same method: each theorem is reduced to an identity in coordinates,

$$(\text{conclusion}) \cdot (\text{non-degenerate factor}) \;=\; \sum_i c_i \cdot (\text{hypothesis}_i),$$

with the multipliers c_i written in the source, closed by a decision procedure for commutative rings. We call such an identity a *certificate*. The method is classical [5]; what a formalisation adds is a translation layer to standard Mathlib notions, and an exact account of which non-degeneracy hypotheses are needed and where.

Contributions.

- *Feuerbach's theorem* for a triangle in an arbitrary real inner product space (the plane hypothesis of the headline statement is present but unused), with tangency in Mathlib's sense, including the degenerate equilateral case; Euler's formula $OI^2 = R(R-2r)$, the excentral formula $OI_i^2 = R(R+2r_i)$, Euler's inequality $2r \le R$, and the distances $|NI| = R/2 - r$, $|NI_i| = R/2 + r_i$ (Section 3). This is *not* the first Lean proof of Feuerbach's theorem (Section 8).

- *Miquel's theorem* in $\mathbb{C}$, on the extended sides, under exactly the hypotheses that make the three circles exist and be unique, in a "for every circle through these three points" form and in a form that constructs the three centres (Section 4). This is *not* the first formal proof of Miquel's theorem; it is the first in Lean that we know of, and its hypotheses are weaker than those of the Isabelle version (Section 8).

- *Brahmagupta's formula* for convex cyclic quadrilaterals in EuclideanSpace $\mathbb{R}$ (Fin 2), stated with Mathlib's `Cospherical` and `Sbtw`, together with the diagonal Bretschneider identity for every quadrilateral, the inequality $K^2 \le (s-a)(s-b)(s-c)(s-d)$ for every quadrilateral, and

the equivalence of equality with Ptolemy's equality (Section 5). The area is the shoelace value (Section 10). This is *not* the first Lean proof of Brahmagupta's formula: a complete proof, by essentially the same route, is in the ShadowBench repository (Section 8). The only parts we did not find there are the two all-four-point statements, the inequality and the Ptolemy equivalence.

- The *Malfatti circles* for every triangle in EuclideanSpace $\mathbb{R}$ (Fin 2): existence of three circles of positive radius, circle i tangent (Mathlib's `IsTangent`) to the two side lines through vertex i with its centre strictly on the interior side of each, any two externally tangent (`IsExtTangent`), and Malfatti's radius formula (Section 6). This is the classical construction, not Malfatti's maximal-area conjecture, which is false.

- The verification record of the four developments: gate, tamper tests, the house battery, non-vacuity witnesses, definition bridges and, for all four, a cross-platform rebuild on arm64/macOS (Section 9), together with the limits of these checks; and, in version 2, Mathlib-only restatements with non-degenerate witnesses (Layer 14) and a comparison with independently written classical statements (Layer 15, Section 9.1).

Why a paper and not only Mathlib pull requests. The code alone does not record three things that this paper does. First, the hypothesis analysis: which non-degeneracy conditions each statement needs and why, how they compare with earlier formalisations (Miquel against the Isabelle version, Feuerbach's equilateral case), and which hypotheses turned out to be unused. Second, the passage from a polynomial certificate to a usable geometric statement in Mathlib's vocabulary, which is where most of the work and most of the design choices lie (the coincident-intersection branch of Miquel, the quarter-angle parametrisation of Malfatti, shoelace area for Brahmagupta). Third, the verification methodology together with an honest account of its limits, including what the mutation and satisfiability layers did *not* cover. The developments themselves remain candidates for Mathlib; for Feuerbach that route runs through Wang's open pull request. In short, of the four theorems only the Malfatti construction is new as a formalisation; Feuerbach and Brahmagupta are second Lean proofs, and Miquel is a further formal proof distinguished by its hypotheses.

1.1 Availability

All four developments are published:

- `Feuerbach`: `https://github.com/jpbald93/feuerbach`, commit `2bf41901548aed8b0e43d2f1bd10bb1a1d54ebf8`;

- `Miquel`: `https://github.com/jpbald93/miquel`, commit `401de13c0c4e51f67f87f7252f147048084c4a01`;

- `Brahmagupta`: `https://github.com/jpbald93/brahmagupta`, commit `c5c3ce014dc450ca212df76fade8948aa45421e0`;

- `Malfatti`: `https://github.com/jpbald93/malfatti`, commit `d39519346e71c9af8c349490a4dac5388de4d8f2`.

The code in all four repositories (Lean sources, gate and tamper scripts, and the numerical checkers) is released under the Apache License 2.0, the licence used by Mathlib; this paper is released under CC BY 4.0. The line references in Table 1 are links to these pinned commits. For all four

repositories every `.lean` file at these commits is byte-identical to its counterpart in the copies from which this paper was written and on which the checks of Section 9 were run. (Those working copies additionally hold one unpublished scratch file each, `scratch/Ax*.lean`, used only to print axiom reports; it is cited nowhere.) The toolchain is Lean `v4.33.1` [7] with Mathlib [9] revision `0df444a3` (tag `v4.33.1`), pinned in each `lake-manifest.json`. The libraries, counted on those copies, are `Feuerbach` (five files and an import-only root file, 630 lines in total), `Miquel` (six files and a root, 1,036 lines, of which 346 are the definition bridges), `Brahmagupta` (four files and a root, 363 lines) and `Malfatti` (five files and a root, 984 lines, of which 202 are the definition bridges and 277 the non-vacuity witnesses). All declarations are in the namespace of the library name. Supporting material (the Layer-14 and Layer-15 reports, the blind writer's statements and report, the transcript audit, and the reviews of this version with the fixes made in response) is in the folder `PaperB-feuerbach-miquel-brahmagupta-malfatti` of the repository `https://github.com/jpbald93/LEAN-verification`.

Version 2. Each repository has a new published commit (Feuerbach `83d2486e8c67c26e9642550be397652f72f16850`, Miquel `9f99fbf4ebc027ea5861abe11416e55bde85a7a6`, Brahmagupta `ded0ff1c47909479590b1d13ccba1d0d5d14ba26`, Malfatti `c0f16c5521c524eba5a787e68e7568fde738208f`) that adds the Layer-14 and Layer-15 files of Section 9.1: `Feuerbach/L14.lean` (136 lines) and `Feuerbach/L15Classical.lean` (233), `Miquel/L14.lean` (168) and `Miquel/L15Classical.lean` (139), `Brahmagupta/L14.lean` (96) and `Brahmagupta/L15Classical.lean` (282), `Malfatti/L14.lean` (138) and `Malfatti/L15Classical.lean` (239). The same commit adds their imports to the root file, extends the declaration list of `gate.sh` and updates the `README.md`. The Layer-15 declarations are in the namespace `<Lib>.Classical`, for example `Feuerbach.Classical`. All earlier library files are byte-identical (SHA-256) to those at the commits above, so every statement, line reference and count given for those commits is unchanged. The pins above, and the source links, cover version 1; the new files are published at the version 2 commits.

Table 1: Informal statements and their formal counterparts. Line numbers refer to the first line of the declaration. For Feuerbach, t is an `Affine.Triangle` in a real inner product space (no dimension hypothesis unless stated).

Informal statement	Lean name	Source	Hypotheses
Feuerbach (Section 3)			
Circumradius relation $4\,\mathrm{heron} \cdot R^2 = a^2b^2c^2$	`Feuerbach.circ_relation`	`Algebra`:22	three equidistance equations
Certificate $\lvert N-I\rvert^2 = (R/2-kT)^2$	`Feuerbach.nine_in`	`Algebra`:28	frame equations, $k^2 = \mathrm{heron}$, $2kR = abc$, $T(a+b+c)=1$
$h_i^2 \cdot (\text{opposite side})^2 = \mathrm{heron}$	`Feuerbach.height_sq_mul_sq`	`Geometry`:43	t
Euler: $OI^2 = R(R-2r)$	`Feuerbach.dist_circumcenter_incenter_sq`	`Main`:100	t
Excentral Euler: $OI_i^2 = R(R+2r_i)$	`Feuerbach.dist_circumcenter_excenter_sq`	`Main`:105	t, i
Euler's inequality $2r \le R$	`Feuerbach.two_mul_inradius_le_circumradius`	`Main`:111	t
$\lvert NI\rvert = R/2 - r$	`Feuerbach.dist_ninePointCircle_center_incenter`	`Main`:119	t

Informal statement	Lean name	Source	Hypotheses
$\lvert NI_i\rvert = R/2 + r_i$	`Feuerbach.dist_ninePointCircle_center_excenter`	`Main`:126	t, i
Incircle internally tangent	`Feuerbach.isIntTangent_insphere_ninePointCircle`	`Main`:143	t
Excircles externally tangent	`Feuerbach.isExtTangent_exsphere_ninePointCircle`	`Main`:154	t, i
Feuerbach's theorem (Thm. 1)	`Feuerbach.feuerbach`	`Main`:163	t; `Fact (finrank ℝ V = 2)` (unused)
Non-vacuity on the 3-4-5 triangle	`Feuerbach.feuerbach_nonvacuous`	`Main`:171	none
Bridge: qf $= \lVert pu + qv\rVert^2$	`Feuerbach.qf_eq_norm_sq`	`Bridge`:26	none
Bridge: heron $= (bc\sin A)^2$	`Feuerbach.heron_eq_sq_two_mul_area`	`Bridge`:36	none
Miquel (Section 4); points in $\mathbb{C}$			
Orientation Dp	`Miquel.Dp`	`Frame`:26	definition
Concyclicity determinant Q	`Miquel.Q`	`Basic`:23	definition
Certificates (x- and y-part)	`Miquel.cert_x`, `Miquel.cert_y`	`Coeffs`:91, `Coeffs`:98	none
Tangency and separation identities	`Miquel.tang`, `Miquel.sep`	`Coeffs`:106, `Coeffs`:113	none
Miquel in the frame	`Miquel.frame`	`Frame`:167	$\Delta \neq 0$, $n \neq 0, 1$, $t \neq 0$, $u \neq 0$
Miquel's theorem (Thm. 2)	`Miquel.miquel`	`Main`:54	$\mathrm{Dp}(A, B, C) \neq 0$; $n, t, u \notin \{0, 1\}$
Miquel, concyclic form	`Miquel.miquel_concyclic`	`Main`:97	same
Example (midpoints)	`Miquel.miquel_example`	`Main`:136	none
Negative control	`Miquel.miquel_negative_control`	`Main`:155	none
All ten hypotheses met	`Miquel.miquel_hyps_example`	`Witness`:27	none
Witness for `core`: second intersection is $(1, 1)$	`Miquel.core_example_point`	`Witness`:111	none
Bridge: $\mathrm{Dp} = 0 \iff$ `Collinear`	`Miquel.Dp_eq_zero_iff_collinear`	`Bridge`:44	none
Bridge: $Q = \mathrm{Dp}\cdot$ `Sphere.power`	`Miquel.Q_eq_Dp_mul_power`	`Bridge`:92	$\lvert R - O\rvert = \lvert S - O\rvert = \lvert P - O\rvert$
Brahmagupta (Section 5); points in EuclideanSpace $\mathbb{R}$ (Fin 2)			
Signed (shoelace) area	`Brahmagupta.signedArea`	`Basic`:25	definition
Area $K = \lvert$signed area$\rvert$	`Brahmagupta.area`	`Basic`:29	definition
Equals the shoelace sum	`Brahmagupta.signedArea_eq_shoelace`	`Basic`:31	none
Diagonal Bretschneider $16K^2 = 4p^2q^2 - (b^2 + d^2 - a^2 - c^2)^2$	`Brahmagupta.sixteen_area_sq`	`Basic`:47	none (any 4 points)
Algebraic identity	`Brahmagupta.brahmagupta_identity`	`Basic`:55	none
$K^2 \leq \prod(s - \cdot)$	`Brahmagupta.area_sq_le`	`Basic`:65	none (any 4 points)
$K^2 = \prod(s - \cdot) \iff pq = ac + bd$	`Brahmagupta.area_sq_eq_iff_ptolemy`	`Cyclic`:21	none (any 4 points)
Brahmagupta, squared	`Brahmagupta.brahmagupta_sq`	`Cyclic`:47	`Cospherical`; diagonals cross (`Sbtw`)

Informal statement	Lean name	Source	Hypotheses
Brahmagupta's formula (Thm. 3)	`Brahmagupta.brahmagupta`	`Cyclic`:63	same
Contrapositive test for cyclicity	`Brahmagupta.not_cospherical_of_area_sq_ne`	`Cyclic`:76	diagonals cross; $K^2 \neq \prod(s - \cdot)$
Unit square: all hypotheses	`Brahmagupta.unitSquare_example`	`Examples`:66	none
Rhombus is not cyclic	`Brahmagupta.rhombus_not_cospherical`	`Examples`:100	none
Bridge: $K = \frac{1}{2}\,pq\sin\theta$	`Brahmagupta.area_eq_half_mul_sin_angle`	`Bridge`:28	none
Malfatti (Section 6); t a triangle; headlines in EuclideanSpace $\mathbb{R}$ (Fin 2)			
Tangent relation $K(p,q,w) = 0$ from Heron	`Malfatti.K_eq_zero_of_heron`	`Algebra`:31	$r > 0$; $p, q, w \in (0,1)$; Heron
Radius formula, parametrised	`Malfatti.radius_eq`	`Algebra`:52	$p \in (0,1)$, $q, w > 0$, $K = 0$ ($r > 0$ unused)
Pairwise tangency identity	`Malfatti.dist_identity`	`Algebra`:69	$p, q > 0$, $K = 0$ ($p < 1$ unused)
Heron $r^2 s = (s-a)(s-b)(s-c)$	`Malfatti.heron`	`Geometry`:134	t; `finrank ℝ V = 2`
The circles (definition)	`Malfatti.malfattiCircle`	`Geometry`:34	definition
Side tangency, interior side	`Malfatti.malfattiCircle_tangent_side`	`Geometry`:240	t; `finrank ℝ V = 2`; i, j, k distinct
Pairwise external tangency	`Malfatti.malfattiCircle_isExtTangent`	`Geometry`:280	t; `finrank ℝ V = 2`; $i \neq j$
Radius formula	`Malfatti.malfattiCircle_radius_formula`	`Geometry`:298	t; `finrank ℝ V = 2`
Existence (Thm. 4)	`Malfatti.malfatti_circles_exist`	`Main`:15	t
Existence with radius formula	`Malfatti.malfatti_circles_formula`	`Main`:31	t
Instance at $(0,0), (1,0), (0,1)$	`Malfatti.malfatti_circles_formula_triW`	`Witness`:97	none
Bridge: $K(\tan a, \tan b, \tan c) = 0 \iff a + b + c = \pi/4$	`Malfatti.K_bridge`	`Bridge`:50	$a, b, c \in (0, \pi/4)$
Bridge: $D(r, \tan\theta) = r/\sin 2\theta$	`Malfatti.D_bridge`	`Bridge`:38	$\theta \in (0, \pi/2)$
Bridge: tparam $= \tan(A_i/4)$	`Malfatti.tparam_bridge`	`Bridge`:115	t; `finrank ℝ V = 2`
Bridge: $r\mu(\ldots)$ is the classical radius	`Malfatti.mu_bridge`	`Bridge`:162	t; `finrank ℝ V = 2`

2 The method

The four proofs follow the three steps of [10].

1. *Coordinates.* Miquel's theorem is stated for points of $\mathbb{C}$ and Brahmagupta's for points of EuclideanSpace $\mathbb{R}$ (Fin 2); in both, the real and imaginary parts (or the two coordinates) are the variables. Feuerbach's theorem is stated for an `Affine.Triangle` in an arbitrary real inner product space, and no isometry to $\mathbb{R}^2$ or $\mathbb{C}$ is used. Instead the argument works in the affine frame $(p_0;\, p_1 - p_0,\, p_2 - p_0)$ of the triangle, with the three side lengths a, b, c as

real parameters: every Gram entry is a polynomial in a, b, c, and square roots never appear because the side lengths are themselves variables. The Malfatti development likewise works with a triangle in a general 2-dimensional real inner product space; its variables are the inradius r and the tangents p, q, w of the quarter angles, so that every length is a rational function of them.

2. *Certificate.* After clearing denominators the conclusion is an explicit combination of hypothesis polynomials. The multipliers are computed outside Lean (with SymPy) and written into the source; Lean checks them with `ring` or `linear_combination`. No proof uses `native_decide` or `polyrith`; `decide` appears only for finite facts about Fin 3.

3. *Translation.* The coordinate statement is connected with Mathlib's notions: `EuclideanGeometry.Sphere.IsIntTangent`, `EuclideanGeometry.Sphere.IsExtTangent`, `Affine.Simplex.circumcenter`, `Affine.Simplex.incenter` and `Affine.Simplex.excenter` for Feuerbach; `dist` for Miquel, with bridges to `Collinear` and `EuclideanGeometry.Sphere.power`; `EuclideanGeometry.Cospherical`, `Sbtw` and Mathlib's Ptolemy theorem for Brahmagupta; `Affine.Simplex.incenter`, `Affine.Simplex.inradius`, `Affine.Simplex.touchpoint`, `EuclideanGeometry.Sphere.IsTangent`, `EuclideanGeometry.Sphere.IsExtTangent` and `AffineSubspace.SSameSide` for Malfatti.

The certificates differ greatly in size. Brahmagupta's are closed by `ring` in a few lines. Miquel's are linear in the unknown point but of moderate size. Feuerbach's `nine_*` and `euler_*` certificates are long (a few hundred monomials each) but each compiles in under a second; a clean build of the whole Feuerbach library took 14 s on the build machine.

3 Feuerbach's theorem

Statement. Let V be a real inner product space, P a metric affine space over V, and t an `Affine.Triangle` $\mathbb{R}$ P (three affinely independent points). Mathlib defines the circumcentre O, circumradius R, incentre I and inradius r, the excentres I_i and exradii r_i (as `t.excenter {i}`, `t.exradius {i}`), the insphere, exspheres and the nine-point circle with centre N, and the predicates `IsIntTangent` and `IsExtTangent` for spheres.

Theorem 1 (`Feuerbach.feuerbach`, `Main`:163)**.** *For every triangle t, the incircle of t is internally tangent to the nine-point circle, and for each $i \in$ Fin 3 the excircle opposite vertex i is externally tangent to the nine-point circle:*

```
theorem feuerbach [Fact (Module.finrank ℝ V = 2)] (t : Triangle ℝ P) :
    t.insphere.IsIntTangent t.ninePointCircle ∧
      ∀ i : Fin 3, (t.exsphere {i}).IsExtTangent t.ninePointCircle
```

The two conjuncts are proved separately as `Feuerbach.isIntTangent_insphere_ninePointCircle` and `Feuerbach.isExtTangent_exsphere_ninePointCircle`, with *no* dimension hypothesis: everything the argument uses lies in the plane of the triangle. The hypothesis `Fact (finrank ℝ V = 2)` appears in `Feuerbach.feuerbach` only because it is part of the target statement, and it is unused. Non-degeneracy is free, since an `Affine.Triangle` is affinely independent by definition. The only extra ingredient for the internal case is `Nontrivial V`, which `Feuerbach.nontrivial_of_triangle` derives from the triangle.

The equilateral case. For an equilateral triangle $R/2 - r = 0$, so the incircle and the nine-point circle coincide. Mathlib's `IsIntTangent` includes this degenerate same-sphere case, and the proof goes through `EuclideanGeometry.Sphere.isIntTangent_iff_dist_center`, which handles it; no case split is needed. The numerical script confirms $|NI| = 0 = R/2 - r$ for an equilateral triangle.

Euler's formula and inequality. On the way we prove

$$OI^2 = R(R - 2r), \qquad OI_i^2 = R(R + 2r_i), \qquad 2r \le R$$

(`Feuerbach.dist_circumcenter_incenter_sq`, `Feuerbach.dist_circumcenter_excenter_sq`, `Feuerbach.two_mul_inradius_le_circumradius`, `Main`:100–111). A search of the pinned Mathlib for `inradius` together with `circumradius` found neither Euler's formula nor the inequality, so they are new relative to that revision; Wang's repository and pull request (Section 8) also prove versions of them.

Proof architecture. The development has three layers.

- `Feuerbach/Algebra.lean` is pure real algebra. With $u = p_1 - p_0$, $v = p_2 - p_0$ and side lengths $a = |p_1p_2|$, $b = |p_0p_2|$, $c = |p_0p_1|$, the quadratic form $\mathrm{qf}(a, b, c, p, q) = p^2c^2 + pq(b^2 + c^2 - a^2) + q^2b^2$ is $\|pu + qv\|^2$, and $\mathrm{heron}(a, b, c) = (2a^2b^2 + 2b^2c^2 + 2c^2a^2 - a^4 - b^4 - c^4)/4$ is the square of twice the area. The circumcentre is $p_0 + xu + yv$; k is twice the area; $T = 1/(\pm a \pm b \pm c)$, so that kT is the in- or exradius. The lemmas `Feuerbach.circ_relation` ($4\,\mathrm{heron} \cdot R^2 = a^2b^2c^2$), `Feuerbach.nine_in`, `Feuerbach.nine_ex0`–`2` (giving $|N - I_\varepsilon|^2 = (R/2 \mp kT)^2$) and `Feuerbach.euler_in`, `Feuerbach.euler_ex0`–`2` (giving $|O - I_\varepsilon|^2 = R^2 \mp 2R\,kT$) are each one `linear_combination`. The multipliers were generated by solving the T-free homogenised identity by linear algebra over an ansatz and adding the multiple of $T \cdot D - 1$ by hand.

- `Feuerbach/Geometry.lean` connects Mathlib's definitions with this algebra. The key step is `Feuerbach.height_sq_mul_sq`: $h_i^2 \cdot (\text{opposite side})^2 = \mathrm{heron}$, a Gram-determinant identity (`Feuerbach.gram_of_foot`) applied at the altitude foot. It turns Mathlib's excentre weights, which are $\pm 1/h_i$, into weights proportional to $\pm$ the side lengths. `Feuerbach.frame_data` expresses O and N in the frame and derives $2kR = abc$; `Feuerbach.excenter_data` gives the excentre coordinates and $r_\varepsilon = k/(\pm a \pm b \pm c)$, taking positivity of the weight sum from Mathlib.

- `Feuerbach/Main.lean` instantiates a generic lemma `Feuerbach.core` at the incentre and the three excentres. Euler's formula gives $2r \le R$, which fixes the sign of the square root and yields $|NI| = R/2 - r$; then Mathlib's `EuclideanGeometry.Sphere.isIntTangent_iff_dist_center` and `EuclideanGeometry.Sphere.isExtTangent_iff_dist_center` give tangency.

Scope. Only the three "true" excircles, with sign set $\{i\}$, are covered. For a triangle the sign sets of size two and three give the same spheres as those of size one and zero; Wang's version is stated for every sign set other than $\emptyset$ and the full set, a mild generalisation that we have not added.

Numerical check. The script `code/check_feuerbach.py` (not part of the proof) computes O, H, $N = (O + H)/2$, I, I_a, I_b, I_c from first principles and checks the four distance identities: on 19,998 random triangles the worst relative error is 9.97×10^{-8} (for badly conditioned triangles); isosceles and equilateral cases give errors of order 10^{-15}; exact SymPy checks on the 3-4-5 triangle,

the 13-14-15 triangle and the triangle $(0,0),(3,0),(1,2)$ give residual 0 for the incircle and all three excircles.

4 Miquel's theorem

Model. The plane is $\mathbb{C}$ with Mathlib's `dist`. For points a, b, c let $\mathrm{Dp}(a,b,c)$ (`Miquel.Dp`) be minus the cross product of $b-a$ and $c-a$, so that $\mathrm{Dp}(a,b,c) \neq 0$ exactly when a, b, c are not collinear (`Miquel.Dp_eq_zero_iff_collinear`). Points on the sides are given by real parameters, and the *extended* sides are allowed:

$$F = A + n(B-A), \qquad E = A + t(C-A), \qquad D = B + u(C-B).$$

The hypotheses $n, t, u \notin \{0,1\}$ say that D, E, F are not vertices; together with $\mathrm{Dp}(A,B,C) \neq 0$ they are exactly what makes the triples (A,E,F), (B,F,D), (C,D,E) non-collinear, so that each of the three circles exists and is unique. (The library states this non-collinearity only in prose. Version 2 proves it, for points of the plane on the side lines and distinct from the vertices, as `Miquel.Classical.cov_B2_obligation`; see Section 9.1.)

Theorem 2 (`Miquel.miquel`, `Main`:54)**.** *Under these hypotheses there is $M \in \mathbb{C}$ such that, for every $O \in \mathbb{C}$ and $r \in \mathbb{R}$: if A, E, F are at distance r from O then so is M; if B, F, D are at distance r from O then so is M; and if C, D, E are at distance r from O then so is M.*

The "for every circle" form is meaningful only because the three circles exist; the companion `Miquel.miquel_concyclic` (`Main`:97) states, under the same hypotheses, that there are O_1, O_2, O_3, M with E, F, M on the circle through A centred at O_1, F, D, M on the circle through B centred at O_2, and D, E, M on the circle through C centred at O_3. Miquel's theorem has earlier formal proofs, in Isabelle/HOL and in Coq; what distinguishes this statement is that it assumes only the non-degeneracy conditions above (Section 8).

Certificate. In a translation-invariant frame with $B = A + w$, $C = A + c$ and $M = F + (x, y)$, the three circle conditions are polynomials q_1, q_2, q_3 in (x, y) (`Miquel.q1`, `Miquel.q2`, `Miquel.q3`). A plain identity $q_3 = P\,q_1 + Q\,q_2$ with linear P, Q is impossible: q_1 and q_2 have no constant term, since both circles pass through F, while q_3 does. Multiplying by x or y repairs this. With $K = n\,t\,u\,(n-1)\,|w|^2$ (`Miquel.KK`) the certificates are

$$K\,x\,q_3 = P_1 q_1 + P_2 q_2, \qquad K\,y\,q_3 = P_1' q_1 + P_2' q_2$$

(`Miquel.cert_x`, `Miquel.cert_y`), with P_i, P_i' linear in (x, y) and computed by SymPy. Two further identities complete the argument: `Miquel.tang` handles the case $M = F$ (the constant term of q_3 is a multiple of the cross product of the linear parts of q_1, q_2), and `Miquel.sep` shows that the radical-axis normal is non-zero. All are closed by `ring`. The second intersection point of the first two circles is given explicitly (`Miquel.rpx`, `Miquel.rpy`); `Miquel.rp_on` shows that it lies on both, and the abstract lemma `Miquel.core` uses the certificates to put it on the third, giving `Miquel.frame`. The lemma `Miquel.bridge` identifies the concyclicity determinant `Miquel.Q` and Dp with the frame polynomials, again by `ring`.

Examples. `Miquel.miquel_example` applies the theorem to $A = 0$, $B = 4$, $C = 1 + 3i$ with D, E, F the midpoints, discharging all hypotheses. `Miquel.miquel_negative_control` shows that in the same configuration F, which lies on the first two circles, lies on no circle through C, D, E, so membership of the third circle is a real constraint; numerically F is at distance about 0.94 from that circle.

Numerical check. `code/check_miquel_formal.py` checks the exact formal configuration, extended sides included, on 20,000 random cases: the worst relative residual is 2.3×10^{-11}, while the negative control has residual at least 2.7×10^{-5}.

5 Brahmagupta's formula

Area. Points are elements of `Pt := EuclideanSpace` $\mathbb{R}$ `(Fin 2)`. For a quadrilateral $ABCD$ we define

$$\begin{aligned} \text{signedArea}(A,B,C,D) &= \tfrac{1}{2}((C_0 - A_0)(D_1 - B_1) - (C_1 - A_1)(D_0 - B_0)), \\ K = \text{area}(A,B,C,D) &= |\text{signedArea}(A,B,C,D)|, \end{aligned}$$

half the cross product of the diagonals; `Brahmagupta.signedArea_eq_shoelace` proves that this equals the shoelace sum $\frac{1}{2}\sum(x_i y_{i+1} - x_{i+1} y_i)$. For a simple (non-self-intersecting) quadrilateral this is the enclosed area. For a crossed quadrilateral it is the difference of the two lobes, which is why the main theorems assume convexity (Section 10).

Three steps. Write $a = AB$, $b = BC$, $c = CD$, $d = DA$, $p = AC$, $q = BD$ and $s = (a + b + c + d)/2$.

1. For *every* quadrilateral, $16K^2 = 4p^2q^2 - (b^2 + d^2 - a^2 - c^2)^2$ (`Brahmagupta.sixteen_area_sq`): the diagonal form of Bretschneider's formula, proved by `ring` after expanding distances into coordinates.

2. As a polynomial identity, $16(s-a)(s-b)(s-c)(s-d) = 4(ac+bd)^2 - (b^2 + d^2 - a^2 - c^2)^2$ (`Brahmagupta.brahmagupta_identity`).

3. Hence, for every quadrilateral, $K^2 = (s-a)(s-b)(s-c)(s-d)$ holds if and only if Ptolemy's equality $pq = ac + bd$ holds (`Brahmagupta.area_sq_eq_iff_ptolemy`), and Ptolemy's inequality in Mathlib (`EuclideanGeometry.mul_dist_le_mul_dist_add_mul_dist`) gives $K^2 \le (s-a)(s-b)(s-c)(s-d)$ for every quadrilateral (`Brahmagupta.area_sq_le`).

Mathlib's Ptolemy theorem `EuclideanGeometry.mul_dist_add_mul_dist_eq_mul_dist_of_cospherical` supplies the equality for a convex cyclic quadrilateral.

Theorem 3 (`Brahmagupta.brahmagupta`, `Cyclic`:63)**.** *Let A, B, C, D, P be points of the Euclidean plane with $\{A, B, C, D\}$ `Cospherical`, P strictly between A and C, and P strictly between B and D. Then $K = \sqrt{(s-a)(s-b)(s-c)(s-d)}$.*

`Brahmagupta.brahmagupta_sq` is the squared form. The hypothesis $\text{Sbtw}\, A\, P\, C \wedge \text{Sbtw}\, B\, P\, D$ says that the diagonals cross at a point interior to both; it is what makes Mathlib's Ptolemy theorem apply. `Sbtw` forces $A \neq C$ and $B \neq D$, but *not* that the four vertices are distinct, so the hypotheses do not by themselves describe a non-degenerate convex quadrilateral. For example, $A = B = (-1, 0)$, $C = D = (1, 0)$, $P = (0, 0)$ satisfies all three hypotheses (the points lie on the unit circle and P is strictly between A and C and between B and D), yet the "quadrilateral" has collapsed to a segment. The theorem remains true there, and trivially so: the shoelace area is 0, the sides are $0, 2, 0, 2$, $s = 2$ and $(s-a)(s-b)(s-c)(s-d) = 0$. Such configurations are therefore admitted rather than excluded; they are not counterexamples. When, in addition, the four vertices are pairwise distinct, the two strict-between conditions do make $ABCD$ a convex quadrilateral with the vertices in this order, all four sides are positive, and informally the shoelace value is then

its enclosed area; this last identification is not proved in Lean (Section 10). We have not added a corollary with an explicit distinctness hypothesis, since it would assume more and conclude nothing new.

Relation to the earlier Lean proof. Theorem 3 is not new. The ShadowBench repository has a complete Lean 4 proof, `brahmagupta_formula`, of the same statement (Section 8, [14]), by essentially the same route: diagonals crossing, Mathlib's Ptolemy theorem, a diagonal identity for the squared area proved by `ring` for arbitrary points, and the semiperimeter algebra. Its hypotheses are expressed differently (an explicit centre and radius instead of `Cospherical`; four orientation signs instead of crossing diagonals; the area as the sum of two triangle areas, and K supplied through a hypothesis $K = \text{quadrilateralArea}\ldots$), but these are presentational differences, not mathematical ones. What our development adds beyond that proof is narrow: `Brahmagupta.area_sq_le` and `Brahmagupta.area_sq_eq_iff_ptolemy`, which hold for *any* four points with no cyclicity or convexity hypothesis, and the contrapositive test `Brahmagupta.not_cospherical_of_area_sq_ne` derived from them, for which we found no earlier formal proof; the statement in Mathlib's `Cospherical/Sbtw` vocabulary; the definition bridge `Brahmagupta.area_eq_half_mul_sin_angle`; the witnesses; and the verification record of Section 9. Version 2 adds the angle form of Bretschneider's formula for convex quadrilaterals (Section 9.1); we did not search for earlier formal proofs of it. We did not compile the ShadowBench proof locally (it targets Lean `v4.29.0`; an attempt under our `v4.33.1` toolchain timed out), so our description of it rests on reading the source, which contains no `sorry`.

Examples. `Brahmagupta.unitSquare_example` shows that the square with vertices $(1,0),(0,1),(-1,0),(0,-1)$ satisfies all hypotheses and has area $2 = (\sqrt{2})^2$; `Brahmagupta.unitSquare_brahmagupta` applies both headline theorems to it. `Brahmagupta.rhombus_fails` shows that the convex rhombus $(1,0),(0,2),(-1,0),(0,-2)$ has $K^2 = 16$ while $(s-a)(s-b)(s-c)(s-d) = 25$, so by `Brahmagupta.not_cospherical_of_area_sq_ne` it is not cyclic (`Brahmagupta.rhombus_not_cospherical`); the cyclic hypothesis cannot be dropped. `code/check_brahmagupta.py` checks the formula on 10,000 random cyclic quadrilaterals (relative error below 10^{-13}) and the general identity on 10,000 random quadrilaterals.

6 The Malfatti circles

Construction, not conjecture. Two different statements carry Malfatti's name, and only the first is formalised here. (i) The *Malfatti circles* of a triangle are three circles, each tangent to the two sides through one vertex, any two of them externally tangent; Malfatti (1803) gave a construction and the radius formula below. (ii) *Malfatti's conjecture* (or problem) asserted that these three circles have the largest total area of any three non-overlapping circles in the triangle. The conjecture is *false*: Lob and Richmond observed that it already fails for the equilateral triangle [8], and Goldberg [6] (1967) showed that the Malfatti circles are never optimal; the greedy arrangement was proved optimal by Zalgaller and Los' [16] (1994). We prove nothing about (ii). The numerical script `code/check_malfatti.py` reproduces the equilateral counterexample (side 1: Malfatti circles of total area 0.315670, incircle plus two corner circles of radius $r/3$ of total area 0.319977), but that is a floating-point check, not a theorem.[1]

[1]The script and its output (`code/check.out`) are in the malfatti repository.

Statement. For a triangle t write r for its inradius, I for its incentre, s for its semiperimeter (`Malfatti.semi`) and $a_i = |A_{i+1}A_{i+2}|$ for the side opposite vertex A_i, indices in Fin 3.

Theorem 4 (`Malfatti.malfatti_circles_exist`, `Malfatti.malfatti_circles_formula`, `Main`:15, `Main`:31)**.** *Let t be a triangle in* EuclideanSpace $\mathbb{R}$ (Fin 2)*. There are spheres (circles) c_0, c_1, c_2 such that*

- *each c_i has positive radius;*
- *for pairwise distinct i, j, k, c_i is tangent (`IsTangent`) to the line A_iA_j, and its centre lies strictly on the same side of that line as A_k (`SSameSide`);*
- *for $i \neq j$, c_i and c_j are externally tangent (`IsExtTangent`).*

Moreover the circles can be chosen with radii

$$\rho_i = \frac{r}{2(s-a_i)}\Big(s - r - \big(|IA_{i+1}| + |IA_{i+2}| - |IA_i|\big)\Big).$$

```
theorem malfatti_circles_exist (t : Triangle ℝ (EuclideanSpace ℝ (Fin 2))) :
    ∃ c : Fin 3 → Sphere (EuclideanSpace ℝ (Fin 2)),
      (∀ i, 0 < (c i).radius) ∧
      (∀ i j k, i ≠ j → i ≠ k → j ≠ k →
        (c i).IsTangent line[ℝ, t.points i, t.points j] ∧
        line[ℝ, t.points i, t.points j].SSameSide (c i).center (t.points k)) ∧
      (∀ i j, i ≠ j → (c i).IsExtTangent (c j))
```

`Malfatti.malfatti_circles_formula` has the same conclusion with the radius equation as an additional first conjunct. Non-degeneracy is again free, since an `Affine.Triangle` is affinely independent. The general-space versions, for a triangle in any real inner product space with `Fact (finrank ℝ V = 2)`, are `Malfatti.malfattiCircle_tangent_side` (which also names the tangency point, lineMap(A_i, T, μ) with T the incircle touch point and $\mu \in (0,1)$, so that it lies on the side segment), `Malfatti.malfattiCircle_isExtTangent` and `Malfatti.malfattiCircle_radius_formula`; `Malfatti.malfattiCircle_radius_lt_inradius` shows $\rho_i < r$.

Construction and certificate. Each circle is the image of the incircle under the homothety at a vertex: the centre is $O_i = A_i + \mu_i(I - A_i)$ and the radius $\mu_i r$ (`Malfatti.malfattiCircle`), so tangency to the two sides through A_i is inherited from the incircle (`Malfatti.isTangentAt_side`). With p, q, w the tangents of the quarter angles at A_i, A_{i+1}, A_{i+2}, defined metrically as $p = (|A_iI| - (s-a_i))/r$ (`Malfatti.tparam`), the ratio is $\mu_i = (1+q)(1+w)/(2(1+p))$ (`Malfatti.mu`). The tangent length and the distance to the incentre are $X(r,p) = r(1-p^2)/(2p)$ and $D(r,p) = r(1+p^2)/(2p)$. The one geometric input is Heron's relation in the form $r^2 s = (s-a)(s-b)(s-c)$ (`Malfatti.heron`), proved from the barycentric description of the incentre and the Gram matrix $\langle A_i - I, A_j - I\rangle = r^2 - x_i x_j$ by a `linear_combination` certificate. Heron's relation forces the polynomial relation

$$K(p,q,w) = 1 - p - q - w - pq - qw - wp + pqw = 0$$

(`Malfatti.K_eq_zero_of_heron`), the tangent form of $A/4 + B/4 + C/4 = \pi/4$; the sign of the cofactor is controlled by $0 < p, q, w < 1$. Pairwise tangency (`Malfatti.dist_identity`) and the radius formula (`Malfatti.radius_eq`) are then polynomial identities whose difference sides are explicit multiples of K; the multipliers were found with SymPy.

Scope. Only existence is proved, by verifying an explicit construction; uniqueness of the Malfatti circles is not. In the files of the first version containment is proved only in part: the tangency points lie on the side segments and the centres lie in the interior half-planes. Version 2 proves full containment: each closed Malfatti disc lies in the closed triangle (`Malfatti.Classical.l15_malfatti_subset`, Section 9.1).

Numerical check. `code/check_malfatti.py` computes the formula radii with centres on the bisectors for 2,000 random triangles: the maximum tangency error (to the sides and pairwise) is 2.6×10^{-10}, while adding 0.01 inside the bracket of the formula gives errors of at least 6.5×10^{-3}.

7 Does the statement mean what it says?

Feuerbach. The statement uses only Mathlib's definitions of the triangle centres, the in-, ex- and nine-point spheres, and tangency; there is nothing local to trust. The two local definitions of the algebra layer are bridged in `Feuerbach/Bridge.lean`: `Feuerbach.qf_eq_norm_sq` shows $\mathrm{qf}(\|v-u\|, \|v\|, \|u\|, p, q) = \|pu+qv\|^2$, and `Feuerbach.heron_eq_sq_two_mul_area` shows that heron of the three side lengths is $(c\,b \sin \angle p_1 p_0 p_2)^2$. The witness triangle `Feuerbach.tri345` is identified by `Feuerbach.tri345_spec` as the 3-4-5 right triangle.

Miquel. The statement uses `dist` and the local Dp. `Miquel/Bridge.lean` has 26 bridge lemmas covering all 25 local definitions: Dp is the orientation determinant and vanishes exactly on `Collinear` triples; Q is the 4×4 concyclicity determinant and equals $\mathrm{Dp}(P, R, S)$ times Mathlib's `EuclideanGeometry.Sphere.power` of the fourth point with respect to the circle through P, R, S (`Miquel.Q_eq_Dp_mul_power`); the frame coefficients are identified with orientations and powers of points; the certificate multipliers, which have no standard meaning, have bridges recording their role; and `Miquel.rpx_rpy_spec` shows that $(\mathrm{rpx}, \mathrm{rpy})$ is the only common zero of the two circle polynomials other than the origin. The docstring of `Miquel.Q` ("vanishes iff concyclic or collinear") was misread by one automated reviewer as a false claim; the intended reading, which is correct, is "all four points lie on a common circle or all four lie on a common line", and the wording could be tightened in a later version.

Brahmagupta. `Brahmagupta.signedArea_spec` shows that the signed area is half the 2×2 determinant of the diagonal vectors and splits along AC into the signed areas of ABC and ACD; `Brahmagupta.area_eq_half_mul_sin_angle` shows $K = \frac{1}{2}\, AC \cdot BD \cdot \sin\theta$ with θ the angle between the diagonals. Neither identifies K with a measure (Section 10).

Malfatti. The headline statements use only Mathlib's triangle, spheres, lines, tangency predicates, `SSameSide`, incentre and inradius, and the local semiperimeter. `Malfatti/Bridge.lean` has nine bridge lemmas, one for each of the nine local definitions: `Malfatti.X_bridge` ($X(r, \tan\theta) = r/\tan 2\theta$, the tangent length $r\cot(A/2)$); `Malfatti.D_bridge` ($D(r, \tan\theta) = r/\sin 2\theta$, the distance from a vertex to the incentre); `Malfatti.K_bridge` ($K(\tan a, \tan b, \tan c) = 0 \iff a + b + c = \pi/4$); `Malfatti.semi_bridge` (half the perimeter, and the sum of the tangent lengths); `Malfatti.tl_bridge` ($s - a_i$, and the distance from the vertex to the incircle touch point); `Malfatti.tparam_bridge` ($\tan(A_i/4)$ with Mathlib's unoriented angle); `Malfatti.mu_bridge` ($r\mu$ at the quarter-angle tangents is Malfatti's classical radius); `Malfatti.ratio_bridge` (the ratio ρ_i/r, and the homothety ratio at the vertex); and `Malfatti.malfattiCircle_bridge` (the defining properties of a Malfatti circle). The bridges hold under stated hypotheses, which we record

exactly: `Malfatti.K_bridge` needs $a, b, c \in (0, \pi/4)$; `Malfatti.D_bridge` needs $\theta \in (0, \pi/2)$; and the geometric bridges (`semi`, `tl`, `tparam`, `mu`, `ratio`, `malfattiCircle`) assume a 2-dimensional plane, `Fact (finrank ℝ V = 2)`. Only `Malfatti.X_bridge` is unconditional. These hypotheses are the natural ranges (quarter angles of a triangle lie in $(0, \pi/4)$), but a reader should not read the bridges as unconditional identities.

8 Related work

Each project began with a prior-art search of Mathlib (sources and the lists `docs/100.yaml` and `docs/1000.yaml`), GitHub code, repository and pull-request search, and other systems. The searches are recorded in the `SPIKE_REPORT.md` of each repository.

Feuerbach: not the first Lean formalisation. Feuerbach's theorem has earlier Lean formalisations, and our contribution is *not* priority. Weiyi Wang (GitHub `wwylele`), the author of Mathlib's `NinePointCircle.lean`, has a standalone repository `wwylele/Feuerbach` [15], created on 29 January 2026, with continuous integration passing and no `sorry` in its 741-line `Feuerbach/Basic.lean`. It targets Lean `v4.28.0-rc1`, follows a vector proof after Scheer [12] using areas, heights, Heron's formula and sines, and proves the internal tangency of the insphere, the external tangency of the exspheres (for every sign set other than $\emptyset$ and the full set), the distance OI and the inequality $2r \leq R$. Wang has also opened Mathlib pull request #44236 [11], "feat(Geometry/Euclidean): prove Feuerbach's theorem", created on 26 September 2026; on 7 October 2026 it was open and labelled `blocked-by-other-PR` and `merge-conflict`. At the pinned Mathlib revision (`v4.33.1`), `docs/100.yaml` has no entry for theorem 29. Outside Lean, Wiedijk's list [17] records formalisations in HOL Light (Harrison), Isabelle (Paulson) and Rocq (the CertiGeo team). Our development was written without consulting Wang's code; we found his repository and pull request only in a final prior-art search, after our proof was complete. It is a second, independent Lean proof by a different route (no angles, sines or Heron import; certificate-driven; about 490 lines for the mathematical content before witnesses and bridges were added), and it should be read as such. Given that #44236 is open, contributing to or comparing with that pull request is the natural route to Mathlib, rather than a competing one.

Independent formalisations in Atlas. Independent Lean formalisations of three of the four theorems were made public in the repository `facebookresearch/atlas-lean` [4]: `FeuerbachTheorem.lean`, `MiquelTheorem.lean` and `BrahmaguptaArea.lean`. All three first appeared on 5 October 2026, between 21:37 and 21:40 UTC, in a batch of import pull requests (#46, #61, #62, #63 and #65; the three files themselves are in #62 and #63). Our Miquel, Malfatti, Feuerbach and Brahmagupta repositories were made public only on 7 October 2026, so *on the public record the Atlas files are earlier than ours.* Our own evidence of earlier completion is private: *working-copy file timestamps* (not commit timestamps) and our run log, which show the Feuerbach algebra file on 28 September 2026 and the Miquel and Brahmagupta certificate and main files complete by 03:45 UTC on 5 October 2026. That is, our working copies were complete before the Atlas import, but our repositories were published after it. We did not consult the Atlas files, and our developments are independent derivations; we claim *no public priority* over them, only independent derivation. (Atlas has no Malfatti file; our claim for Malfatti in the Summary rests on the searches below, not on dates.) The Atlas files are imported from an upstream repository, `fairinternal/formal-math`, which is private; the date on which they were first written there is not public, so their earlier private authorship cannot be excluded either.

Miquel: not the first formalisation. An earlier search by the Miquel project (Mathlib sources and both lists, GitHub, the Archive of Formal Proofs, HOL Light, Coq/Rocq and Mizar) found no formal proof, but it was not repeated before writing and it missed two. The Isabelle Archive of Formal Proofs has the entry *Miquel's Theorem in Isabelle/HOL* by A. Freitas Ramos, D. Barros Hulak and R. J. G. B. de Queiroz, dated 11 August 2026 [2]. It is sorry-free, works in complex coordinates with the points on the side *lines* (the extended sides, as here), and proves both a pivot form (`miquel_pivot`) and a concurrence form (`miquel_concurrent`) in which the common point is constructed, as the reflection of r in the line of centres. Mathematically this is the content of `Miquel.miquel` and `Miquel.miquel_concyclic`, two months earlier. Coq's HighSchoolGeometry library (F. Guilhot) has `Theorem Miquel`, in pivot form, proved with `Qed` (`theories/applications_cocyclicite.v`, line 270) [3]; it is relative to an axiomatised synthetic geometry, but it is a library theorem, not merely an automated-prover benchmark (Miquel's theorem is also a standard benchmark for Wu and Gröbner-basis provers [5]). We therefore claim no priority for Miquel's theorem. The remaining difference is *hypothesis economy*. Our `Miquel.miquel` assumes only $\mathrm{Dp}(A,B,C) \neq 0$ and $n,t,u \notin \{0,1\}$, which say exactly that the triangle is non-degenerate and that no side point is a vertex. The AFP `miquel_concurrent` assumes these conditions too, and in addition that the two given circles have positive radii, $0 < R_1$ and $0 < R_2$, and that the two centres and the common point r are not collinear, $\neg$ collinear $o_1\ o_2\ r$ (which excludes the case where the two circles are tangent at r); its `miquel_pivot` assumes $m \neq p$, $m \neq q$, $m \neq r$, so that the Miquel point is distinct from the three side points. Our proof handles the case $M = F$, which includes the case where the first two circles are tangent at F, by the identity `Miquel.tang`, and it does not require M to differ from the side points, so neither kind of extra hypothesis is needed.

Brahmagupta. In the pinned Mathlib the only `brahmagupta` declaration is the Brahmagupta–Fibonacci identity; `docs/1000.yaml` has an entry "Brahmagupta theorem" that refers to the perpendicular-diagonals theorem, not the area formula, with no declaration; there is no Bretschneider formula and no quadrilateral area. Mathlib does have Ptolemy's theorem and Ptolemy's inequality, on which we build. GitHub searches (Lean, Coq, Isabelle, OCaml; `brahmagupta`, `Bretschneider`, "cyclic quadrilateral", repositories and Mathlib pull requests) found no proof of the area formula, but that search was wrong. The ShadowBench autoformalisation benchmark [13] contains, under `autformalization/problems/geometry/L2/geo_gen_L2_007/`, both skeleton files `docs/skeletons/Skeleton1.lean`–`Skeleton4.lean`, which end in `sorry`, and a *complete* proof in `ShadowBench/Source/Main.lean` [14]. At commit `7541fcecef97cc124f38355a2cd2ba0ec8485dd5` (Lazar Milikic, 12 June 2026) that file has 334 lines, no `sorry`, the main theorem `brahmagupta_formula` (line 319) and the lemma `brahmagupta_area_sq_semiperimeter` (line 290). Its statement is ours: a convex cyclic quadrilateral, the semiperimeter, $K = \sqrt{(s-a)(s-b)(s-c)(s-d)}$, with the area supplied by a bridge hypothesis `h_area : K = quadrilateralArea ...`. An earlier draft of this paper said that the closest ShadowBench results were `sorry` statements; that was false, because we had read only the skeletons. We claim no priority for Brahmagupta's formula. We have not compiled the ShadowBench proof (Section 5). A further search found no proof of the all-four-point statements `Brahmagupta.area_sq_le` and `Brahmagupta.area_sq_eq_iff_ptolemy`. A web and AFP search found no Brahmagupta area formula in Isabelle, HOL Light (whose `100/heron.ml` has Heron's formula only), Coq/Rocq or Mizar. These searches are best-effort only.

Malfatti. The Malfatti project searched, on 7 October 2026, the pinned Mathlib (including `Archive/`, `docs/100.yaml` and `docs/1000.yaml`), the Isabelle Archive of Formal Proofs (a mirror

of 1,045 entries, and the AFP site search), HOL Light, HOL4, GeoCoq, the Rocq 100-theorems collection, the Mizar Mathematical Library, GitHub code and repository search for Lean, Coq, Isabelle, SML and OCaml, and Wiedijk's 100 and 1000 lists, and found no formalisation of the Malfatti circles. The only related item is an open wishlist entry in a Mathlib issue (#6091) for the different statement that the Malfatti circles are never optimal (item (ii) of Section 6), not a proof. The search did not cover the Lean Zulip, and GitHub code search indexes default branches only.

Summary. The Atlas files of 5 October 2026 (above) are independent formalisations, public before our repositories, of Feuerbach, Miquel and Brahmagupta; we claim only independent derivation, not public priority, against them. They do not change the conclusions below. As far as these searches go, we found no earlier formalisation of the Malfatti circle construction; this is the only one of the four theorems we believe to be formalised here for the first time. Our Brahmagupta development is a second Lean proof of a theorem already proved in Lean in the ShadowBench repository; only its all-four-point inequality and Ptolemy equivalence were not found elsewhere. Our Feuerbach development is an independent formalisation of a theorem already formalised in Lean (outside Mathlib, with a Mathlib pull request open) and in three other systems. Our Miquel development is a further formalisation of a theorem already proved in Isabelle/HOL and in Coq, distinguished by its weaker hypotheses. The searches are not exhaustive; both the Feuerbach and the Miquel cases show that search can miss existing work (GitHub code search returned no hits for Wang's repository, the first Miquel search missed an AFP entry, and the first Brahmagupta search saw only the `sorry` skeletons of a benchmark problem whose complete proof sat in the same directory). The claim for Malfatti should be read with this record in mind.

9 Verification

The Feuerbach, Miquel and Brahmagupta results in this section were obtained on 6–7 October 2026, and the Malfatti results on 7 October 2026, on the build machine (x86_64, Linux); some were repeated on a second machine (arm64, macOS), as stated. The working copies are not git checkouts. For all four developments every `.lean` *file at the published commits (Section 1.1) is byte-identical to its counterpart in the checked copies (SHA-256 of every* `.lean` *file, and* `cmp` *on fresh clones). Layers 14 and 15 (Section 9.1) were added in version 2, on 21–22 October 2026.*

What the checks taught us. We lead with the lessons that seem reusable beyond these four libraries; the counts follow, and should be read in their light.

1. *A witness must exercise the certificate, not merely type-check.* The first `Miquel.core` witness took the zero polynomial as its third circle; every check passed, yet it exercised none of the certificate multipliers. It was found by reading and replaced by three genuine circles with a second intersection point proved to be $(1, 1)$, not the origin. Gate, tamper and satisfiability layers all accepted the degenerate witness; none of them can detect this.

2. *A witness should apply the theorem, so that downstream code polices it.* Rewriting the Malfatti witnesses to state the target lemma's conclusion and prove it by applying the lemma killed 12 of 16 genuine mutation survivors, and in Miquel the four survivors in `Witness.lean` are rejected by the declarations that consume them. Witnesses written this way are also what the harness can link to their theorems (the satisfiability table below).

3. *Ablation is informative when a proof survives, not when it breaks.* The surviving deletions found five redundant hypotheses (`_hN` in `Miquel.rp_on` and four in Malfatti), which is a

concrete defect in the statements. A broken proof after deletion says only that this proof used the hypothesis.

What the battery is and is not. The checks below are not thirteen independent safeguards. They separate into four kinds of evidence. *Kernel certification* (build, forbidden-token grep, `#print axioms`, gate) shows that the stated propositions are proved from the standard axioms; the tamper test checks that the gate itself rejects planted defects. *Statement-meaning evidence* (definition bridges, hand-written witnesses, numerical scripts, and the author's reading of Table 1) bears on whether the propositions are the intended ones; only the bridges and witnesses are kernel-checked, and their selection is manual. *Heuristic testing* (mutation, contradiction probes, ablation, satisfiability, bridge inventory) is incomplete by construction: a contradiction probe that finds nothing is not a satisfiability proof, mutants are generated only for some declarations, and timeouts are counted with type errors. *Platform reproducibility* (the arm64 rebuild) checks the toolchain, not the mathematics. The harness also has visible limits: it parsed a docstring as a declaration and generated eight phantom mutants and a phantom satisfiability success from it, and its bridge inventory had to be corrected by hand. Many layers share the same sources and the same author, and their failures are correlated.

Build and axioms. A clean `lake build` of each library completes with 0 errors and 0 warnings; for Feuerbach, Miquel and Brahmagupta this took 8,710, 8,711 and 8,710 jobs before the witness and bridge files were added (on the final sources: 8,712, 8,713, 8,711, and 8,712 for Malfatti). A grep of the library sources for `sorry`, `admit`, `native_decide`, `axiom`, `#eval`, `run_cmd`, `set_option`, `macro`, `elab`, `syntax`, `notation` and `import Lean` is empty, and `#print axioms` reports only `propext`, `Classical.choice` and `Quot.sound`.

House battery. The battery of the earlier AFM papers was run serially on the build machine after the definition bridges and witnesses were added.

- *Gate.* `gate.sh` rejects the forbidden tokens, builds the library, generates its own `#print axioms` query for every declaration of the library (theorems, lemmas and definitions), and fails on any non-standard axiom. Result: PASS, standard axioms only, at **46** declarations (Feuerbach), **84** (Miquel, after the `core` witness was replaced, below), **30** (Brahmagupta) and **77** (Malfatti).

- *Tamper tests.* `tests/tamper.sh` plants six defects in scratch copies (a `sorry`; a split `axiom`; a fake `#eval` report; a macro; a deleted hypothesis; a wrapped extra axiom) and requires each to be rejected. The deleted hypotheses are `hpos` of `Feuerbach.core`, `hn0` of `Miquel.miquel`, `hcyc` of `Brahmagupta.brahmagupta_sq` and `hK` ($K = 0$) of `Malfatti.mu_lt_one`. Result: **6/6** rejected in each of the four repositories.

- *Mutations.* Small edits to conclusions and hypotheses, with the proof kept, must fail to compile. Results: Feuerbach 177/177 rejected, no survivors; Brahmagupta 78/84 rejected, 6 survivors; Miquel 192/199 rejected, 7 survivors; Malfatti 226/238 rejected, 12 survivors, of which 4 are genuine (below). "Rejected" includes failures by heartbeat timeout (3 for Feuerbach, 8 for Miquel, 1 for Malfatti). These totals cover only the declarations for which the harness generated mutants, and several headline statements are not among them. For Feuerbach the harness generated *no* mutants for `Feuerbach.isIntTangent_insphere_ninePointCircle`, `Feuerbach.isExtTangent_exsphere_ninePointCircle` and `Feuerbach.nontrivial_of_`

`triangle`, so "177/177 rejected" says nothing about the two tangency theorems. For Brahmagupta it generated none for `Brahmagupta.brahmagupta`, `Brahmagupta.brahmagupta_sq`, `Brahmagupta.area_sq_le`, `Brahmagupta.area_sq_eq_iff_ptolemy`, `Brahmagupta.brahmagupta_identity` and `Brahmagupta.not_cospherical_of_area_sq_ne`, that is, for every headline statement of Section 5; the 84 Brahmagupta mutants all lie in the auxiliary lemmas and examples. The five Malfatti declarations without mutants are listed below. All six Brahmagupta survivors lie inside the concrete examples and were reproduced and classified harmless by hand (for instance, a "not concyclic" assertion survives almost any small nudge of a point, because landing exactly on a circle is non-generic). *Miquel.* All Miquel battery figures in this section are from the run on the current sources completed on 7 October 2026 (started 17:58 UTC), after the `core` witness was replaced. There are seven survivors. Three are in `Miquel.miquel_negative_control` (`Miquel/Main.lean`) and four in `Miquel/Witness.lean` (two in `Miquel.miquel_hyps_example`, two in `Miquel.core_example`). All seven were reproduced (each mutant compiles with the original proof) and ruled on by hand: all seven are harmless and none is a genuine defect. Each mutated statement is still a true closed numerical fact under an open ($\neq 0$) condition; in the two `Miquel.core_example` slots the side condition reduces to $4 + 4a^2$ and $b^2 + 4$, strictly positive for every real value. For the four survivors in `Witness.lean` the downstream declaration that consumes the witness (`Miquel.miquel_point_example`, or `Miquel.core_example_conclusion` through `Miquel.core`) rejects the mutant with a type mismatch. *Malfatti.* The non-vacuity witnesses were rewritten so that each states the target lemma's conclusion at concrete arguments and is proved by applying the lemma, with its hypotheses discharged inline; this killed 12 of the 16 genuine survivors of the previous run. Of the 12 survivors in the final run, 8 are an artefact of the harness, which parsed a docstring as a declaration. The 4 that remain are benign: the mutants of the library lemmas `Malfatti.fin3_distinct` (one) and `Malfatti.fin3_third` (two) are true but weaker statements about Fin 3, killed downstream where the dropped conjunct is used, and the mutant of `Malfatti.heron_witness` ($1^2 \mapsto 1^3$) is value-preserving, since $1^3 = 1^2$. For five declarations (`Malfatti.isTangentAt_side`, `Malfatti.malfattiCircle_isExtTangent`, `Malfatti.malfattiCircle_tangent_side`, `Malfatti.sSameSide_center`, `Malfatti.touch_mem`) the harness generated no mutants at all (Section 10). The general lesson is that mutation testing is weak on concrete numerical examples and on non-generic negative claims.

- *Contradictory hypotheses.* No hypothesis set was found contradictory (0 in each).

- *Ablation.* Each hypothesis is deleted in turn and the unchanged proof is recompiled. Result, as (proof broke)/(proof survived): Feuerbach 70/0; Miquel 186/1; Brahmagupta 6/0; Malfatti 66/4. The harness labels the first figure "needed", but that is too strong: a broken proof shows only that *this* proof uses the hypothesis, not that the statement is false without it. The second figure is the informative one, since a surviving proof does show the hypothesis redundant. The unused Miquel hypothesis is `_hN` of `Miquel.rp_on`, already named with an underscore. It is genuinely unnecessary: if $\mathrm{rpN} = 0$ then, with Lean's convention $x/0 = 0$, the point $(\mathrm{rpx}, \mathrm{rpy})$ is the origin, where both circle polynomials vanish. The four unused Malfatti hypotheses are $0 < r$ in `Malfatti.radius_eq`, $p < 1$ in `Malfatti.dist_identity`, $q < 1$ in `Malfatti.mu_lt_one` and $i \neq j$ in `Malfatti.fin3_third` (Section 10).

- *Unused-variable linter.* 0 warnings in all four libraries.

- *Satisfiability.* For each theorem declaration, the harness tries to produce a kernel-checked

`example` exhibiting values for all its hypotheses. Since the reviewers' reports, the harness has a separate route for hand-written witnesses: a witness declaration that applies a theorem with its hypotheses discharged is *linked* to that theorem and reported on its own line, never folded into the certified count. The denominator is *all* theorem declarations, including hypothesis-free and closed statements, which are certified trivially; the ratios below are therefore not coverage of hypothesis-bearing statements. Results, separated:

	machine-certified	witness-linked	unresolved	total
Feuerbach	30	0	**13**	43
Miquel	40	0	**19**	59
Brahmagupta	22	0	**4**	26
Malfatti	26	33	**8**	67

For Malfatti, $26 + 33 = 59$ of 67 declarations are credited and 8 are unresolved; the total is 67, not the 68 of an earlier run, because the harness no longer parses a docstring as a declaration. For the other three libraries no witness is written in the form the linker recognises, so the witness-linked count is 0 and the unresolved counts stand as reported: **13** Feuerbach items (algebra and frame lemmas), **19** Miquel items (bridge and specification lemmas together with `Miquel.dist_eq_of_Q_eq_zero`, `Miquel.circ_aux`, `Miquel.core`, `Miquel.bridge`, `Miquel.miquel` and `Miquel.miquel_concyclic`) and **4** Brahmagupta items. Several of these, including every headline statement, are exercised by the hand-written witnesses listed below, but that coverage is established by reading, not by the harness, and we do not count it. `Miquel.rp_zero` is machine-certified. An unresolved item is one for which no satisfying instance was produced automatically; it is not evidence of vacuity, and a certified item is evidence only that the hypotheses are jointly satisfiable.

- *Bridge inventory.* The scanner's heuristic reports Miquel 21 of 25 notions bridged, Brahmagupta 2 of 4, Feuerbach 0 of 3 and Malfatti 1 of 10. These numbers are under-counts: the heuristic does not match the names in the `Bridge.lean` files. The inventory we rely on was therefore corrected by hand: by inspection every mathematical local definition of the four libraries has a bridge lemma (Section 7); for Malfatti that is nine bridges for nine definitions, the tenth definition being the witness triangle. This is a reading, not a machine check.

Non-vacuity witnesses. *Feuerbach:* `Feuerbach.feuerbach_nonvacuous` instantiates every hypothesis of `Feuerbach.feuerbach` with $V = P = \mathbb{C}$ (of real dimension 2) and the triangle `Feuerbach.tri345` with vertices $0, 4, 3i$, whose affine independence is `Feuerbach.tri345_affineIndependent`; `Feuerbach.algebra_hyps_345` shows that the hypotheses of the algebra lemmas are jointly satisfiable ($a, b, c = 5, 3, 4$, $R = 5/2$, $k = 12$), and `Feuerbach.nine_in_345` instantiates `Feuerbach.nine_in` there. *Miquel:* `Miquel.miquel_hyps_example` discharges all ten hypotheses of `Miquel.miquel` for the midpoint configuration, and `Miquel.miquel_point_example` applies the theorem there; further witnesses cover `Miquel.dist_eq_of_Q_eq_zero`, `Miquel.circ_aux`, `Miquel.bridge` and `Miquel.core`. The `Miquel.core` witness `Miquel.core_example` uses three genuine circles, x^2+y^2-2x, x^2+y^2-2y (meeting at the origin and at $(1,1)$) and x^2+y^2-2, with non-zero certificate multipliers; `Miquel.core_example_point` proves that the second intersection point of the first two is $(1,1)$, not the origin, and `Miquel.core_example_conclusion` applies `Miquel.core` to put it on the third circle. (An earlier version of this witness took the zero polynomial as its third circle and so did not exercise the certificates; it has been replaced.) `Miquel.rp_zero` has no hand-written witness; it is harness-certified satisfiable. *Brahmagupta:*

the unit square and the rhombus of Section 5, through `Brahmagupta.unitSquare_brahmagupta`, `Brahmagupta.unitSquare_square_area` and `Brahmagupta.rhombus_not_cosphericalʼ`. *Malfatti:* `Malfatti.malfatti_circles_exist_triW` and `Malfatti.malfatti_circles_formula_triW` instantiate the two headline theorems at the right triangle `Malfatti.triW` with vertices $(0,0),(1,0),(0,1)$, whose affine independence is `Malfatti.triW_affineIndependent`. The algebraic lemmas are instantiated at $r = 1$, $(p,q,w) = (1/3,1/3,1/7)$ and, so that the r-scaling is exercised, at $r = 2$, $(p,q,w) = (1/3,1/5,3/11)$: Heron's relation (`Malfatti.heron_witness`, `Malfatti.heron_witness_r2`) is fed to `Malfatti.K_eq_zero_of_heron`, whose conclusion $K = 0$ is in turn fed to `Malfatti.radius_eq`, `Malfatti.dist_identity` and `Malfatti.mu_lt_one`; `Malfatti.param_spec` is instantiated at $(r,x,d) = (3,4,5)$. Each witness is the target lemma's conclusion at those arguments, so most changes to the lemma or to a numeral break the application (not all: see the value-preserving survivor above).

Cross-platform rebuild on arm64/macOS. All four libraries were rebuilt on a second machine with a different architecture and operating system (Apple M5 Pro, arm64, macOS), with the same Lean and Mathlib versions, from sources transferred by tarball and checked by SHA-256 on both legs. Before the witnesses and bridges were added, the three builds there completed with exit status 0, 0 warnings and the same job counts as on the build machine (8,710, 8,711, 8,710), and `#print axioms` agreed for all 25 declarations then cited. On the final sources all four libraries built there with exit status 0 and 0 warnings (8,712, 8,713, 8,711 and 8,712 jobs), `gate.sh` passed at 46, 84, 30 and 77 declarations, and `tests/tamper.sh` rejected all six defects in each repository. This rebuild uses the same sources, toolchain and author, so it checks platform independence, not independent authorship.

Limits of these checks. None of these checks shows that a formal statement says what its informal counterpart says. The author has checked each entry of Table 1 by hand against the cited Lean declaration and confirms that the formal statement, with the listed hypotheses, is the informal statement in the same row. This is a single reader's judgement, so it was complemented by a blind re-formalisation and, in version 2, by a comparison with classical statements (Section 9.1).

Blind re-formalisation. The paper's text, with every Lean name, declaration reference and repository link removed, was given to a separate language-model writer that had no access to the code. From that text alone it wrote a Lean statement of each of the four theorems. Each blind statement was then compared with the corresponding library theorem by proving, in Lean against the pinned libraries, both implications: blind statement $\Rightarrow$ library statement and library statement $\Rightarrow$ blind statement. Inside these proofs only the premise, the local definitions and general Mathlib lemmas were allowed; a transitive walk over the dependency closure of all eight implications confirms that none of the fourteen headline theorems of the libraries is used. All eight implications were proved, from the standard axioms only, so all four pairs are equivalent. Two points are worth recording. For Feuerbach the blind statement has no hypothesis that the ambient space is two-dimensional; the library theorem does, and the implication from the library statement to the blind one goes through by restricting to the affine span of the triangle, confirming that this hypothesis is removable, as stated in Section 3. For Feuerbach and Malfatti, identifying the blind statement's explicitly defined centres and radii with Mathlib's uses the library's coordinate lemmas for those points, which are definition bridges and say nothing about tangency. The comparison is statement-level: it shows that the prose and the code state the same propositions, not that either is true. The writer was a model, not an independent human formaliser, and it shares the paper's

wording; the check guards against prose-versus-code drift, not against a shared misreading of the mathematics.

9.1 Layers 14 and 15

These two layers were added in version 2 (21–22 October 2026). Each repository gains the files `<Lib>/L14.lean` *and* `<Lib>/L15Classical.lean` *(Section 1.1); every earlier library file is byte-identical to the published commit. With the new files the gates pass, standard axioms only, at 71 (Feuerbach), 101 (Miquel), 50 (Brahmagupta) and 93 (Malfatti) declarations, against 46, 84, 30 and 77 before.* `tests/tamper.sh` *still rejects all six planted defects in each repository, and a grep of each new Layer-15 file for* `sorry`, `admit`, `native_decide`, `axiom`, `set_option`, `macro`, `notation`, `run_cmd`, `#print` *and* `#eval` *is empty. The battery layers of this section (mutation, ablation, satisfiability) and the arm64 rebuild were not repeated on the new files.*

Layer 14: statements in Mathlib's language and non-vacuity. Each of the four numbered theorems (Theorems 1–4) is restated in `<Lib>/L14.lean` with Mathlib-only types and proved from the library theorem through bridges. That is six declarations: one each for Feuerbach and Brahmagupta, and two each for Miquel and Malfatti. A check of each elaborated type, printed with full names, found no constant from the library's namespace. For each theorem a witness proves every hypothesis on an explicit configuration and also proves that the conclusion is non-degenerate:

restatement	witness and distinctness proved
`Feuerbach.l14_feuerbach`: the nine-point circle is given by its defining property, as *any* sphere through the three side midpoints	`Feuerbach.l14w_feuerbach`: triangle $0, 4, 3i$, circle with centre $1 + \frac{3}{4}i$ and radius $\frac{5}{4}$; the incircle and each excircle differ from it, and the four tangent circles are pairwise distinct
`Miquel.l14_miquel`, `Miquel.l14_miquel_cospherical`: Dp $\neq 0$ becomes $\neg$`Collinear`, the parameters become membership of `line[`$\mathbb{R}$`,·,·]` together with "not a vertex" (equivalent hypotheses), and the circles are written with `Cospherical`	`Miquel.l14w_miquel`: $A = 0$, $B = 2$, $C = 2i$, with $F = -2$ and $E = -2i$ on extended sides and $D = 1 + i$; the Miquel point is forced to be $-2 - 2i$, which differs from all six points; the three circles are distinct
`Brahmagupta.l14_brahmagupta`: the area is written as $\frac{1}{2}\,AC \cdot BD \sin\theta$, with θ Mathlib's angle between the diagonals	`Brahmagupta.l14w_brahmagupta`: $(1,0), (0,1), (-1,0), (\frac{3}{5}, -\frac{4}{5})$; the four vertices are distinct, $BC \neq CD$ (so it is not a square), and the area is $\frac{9}{5}$
`Malfatti.l14_malfatti_circles_exist`, `Malfatti.l14_malfatti_circles_formula`: the statement was already Mathlib-only apart from `semi`, which is written out	`Malfatti.l14w_malfatti`: the scalene triangle $(0,0), (154,0), (35,120)$ with sides $154, 169, 125$; *every* family satisfying the conclusion has pairwise distinct centres and circles (radii positive, but not computed in closed form)

No Mathlib-only restatement turned out false, and none needed a hypothesis beyond those of the library theorem. The `Fact (finrank` $\mathbb{R}$ `V = 2)` hypothesis, which Theorem 1 does not use, *is* needed in the defining-property form: in dimension at least three, a large sphere through the three midpoints is not tangent to the incircle. The Brahmagupta shoelace area is still not formally identified with Lebesgue measure (Section 10). The other rows of Table 1 (Euler's formula, the distances, the all-four-point Brahmagupta statements, the general-space Malfatti lemmas) were not restated, because Layer 14 covers only the numbered theorems.

Layer 15: comparison with classical statements. A separate agent, which saw no formalisation, wrote textbook statements of the four theorems (B1–B4), by its own account from memory apart from one web source, the Wikipedia page on the Malfatti circles. It attached citations (Feuerbach 1822 and Coxeter–Greitzer §5.6; Miquel 1838 and Johnson §184; Brahmagupta and Coxeter–Greitzer §3.2; Malfatti 1803 and Stevanović 2003), but it did not consult these sources and their

section references were not checked. An AI writer that saw only those statements, and neither the code nor this paper, wrote 19 Lean statements with Mathlib notions, each ending in `sorry`. Four of them are construction obligations. The writer defined three notions itself: a convex quadrilateral (the diagonals cross in their interiors), the area of a quadrilateral (Lebesgue measure of the convex hull) and a Malfatti triple (closed discs inside the triangle). Its tool-call transcript, and that of the statement collector, were audited: neither read the paper or the code, and the writer made no web access (a transcript audit is in the verification repository). A comparer copied the statements verbatim and then either proved each from the development or classified it. All the proofs depend only on the three standard axioms. They are collected in `<Lib>/L15Classical.lean` (namespace `<Lib>.Classical`), whose gates list every one of them; the writer's definitions are reproduced there verbatim. Of the 19 statements:

- *covered, 13:* all seven Feuerbach statements (internal and external tangency, the two distance forms, the excentre obligation from Mathlib, and the two remarks, newly proved in these libraries in version 2; the equilateral coincidence was already described informally in Section 3); Miquel on open sides and on side lines, and its non-collinearity obligation (new); the Brahmagupta measure obligation (from Mathlib); Malfatti existence, with containment in the triangle (new); and the Malfatti positivity obligation;

- *narrower, 4:* Brahmagupta's formula in both forms and the angle form of Bretschneider's formula (the last not stated in the version-1 files) are proved for the shoelace area (`Brahmagupta.Classical.cov_B3_a_part`, `Brahmagupta.Classical.cov_B3_b_part`, `Brahmagupta.Classical.cov_B3_remark_bretschneider_part`). For the writer's Lebesgue area they are proved only under the explicit hypothesis `AreaBridge` (Lebesgue measure of the convex hull equals the shoelace area for convex quadrilaterals); that identification is missing. Malfatti's radius formula is proved for the constructed triple (`Malfatti.Classical.cov_B4_b_part`). For *every* Malfatti triple it is proved only from the uniqueness statement (`Malfatti.Classical.cov_B4_b_of_unique`); uniqueness itself is missing;

- *out of scope, 1:* uniqueness of the Malfatti circles (`classical_B4_a_unique`), which the paper does not claim. It is not proved;

- *writer statement false as written, 1:* "a convex quadrilateral is cyclic iff $\angle DAB + \angle BCD = \pi$". The writer's convexity admits four collinear points: $A = (0,0)$, $B = (2,0)$, $C = (1,0)$, $D = (\frac{1}{4}, 0)$ has angle sum π but the points are not cospherical (`Brahmagupta.Classical.cov_B3_remark_cyclic_iff_false`). The forward direction is proved for distinct consecutive vertices (`Brahmagupta.Classical.cov_B3_remark_cyclic_iff_part`).

New proofs, not in the files of version 1: (i) a triangle is equilateral iff $R = 2r$ (`Feuerbach.Classical.l15_equilateral_iff`), so for an equilateral triangle the nine-point circle is the incircle (`Feuerbach.Classical.cov_B1_remark_equilateral`), and otherwise the internal point of contact is unique (`Feuerbach.Classical.cov_B1_remark_unique`); (ii) the three Miquel triples are non-collinear (`Miquel.Classical.cov_B2_obligation`); (iii) the angle form of Bretschneider's formula for every convex quadrilateral, with the shoelace area (`Brahmagupta.Classical.l15_bretschneider_shoelace`); (iv) the counterexample and the forward direction above; (v) each closed Malfatti disc lies in the closed triangle (`Malfatti.Classical.l15_closedBall_incenter_subset`, `Malfatti.Classical.l15_malfatti_subset`, used in `Malfatti.Classical.cov_B4_a`). The writer added no hypothesis that the libraries lack. Where the libraries are more general (Feuerbach in any inner product space, Miquel in $\mathbb{C}$, transported to the writer's plane by a Mathlib isometry), the writer's statements are special cases. As with the blind re-formalisation, the writer

was a model, not an independent human formaliser. It was not supplied this paper's wording, but the classical statements it formalised were themselves collected by an agent.

10 Limitations and further work

- *Feuerbach: prior art.* The theorem is already formalised in Lean by Wang (standalone repository and open Mathlib pull request #44236) and in HOL Light, Isabelle and Rocq. Our development is independent but not first. Smaller points: only the three excircles with sign set $\{i\}$ are stated; the tangency point (the Feuerbach point) is not given explicitly, although Mathlib's characterisation produces it inside an existential (version 2 proves that it is unique for a non-equilateral triangle, `Feuerbach.Classical.cov_B1_remark_unique`); and the long machine-generated `linear_combination` certificates would need shortening before they could be offered to Mathlib.

- *Feuerbach: a global instance.* `Feuerbach/Witness.lean` declares a `noncomputable instance : Inhabited (Triangle` $\mathbb{R}$ $\mathbb{C}$`)` so that the automated satisfiability checker has a default triangle. Adding a default instance can in principle affect elaboration elsewhere; the paper's non-vacuity claim rests on `Feuerbach.feuerbach_nonvacuous`, not on the instance. We have decided to keep it in the published version: it adds no axiom and no hypothesis to any theorem, the satisfiability layer uses it as its default triangle, and removing it would change the pinned commit on which every check of Section 9 was run. It is an orphan instance on a Mathlib type, so it would be dropped from any Mathlib submission.

- *Brahmagupta: prior art.* Brahmagupta's formula is already proved in Lean 4, without `sorry`, in the ShadowBench repository (commit of 12 June 2026). Our development is not first; what it adds is the all-four-point inequality and Ptolemy equivalence, the statement in Mathlib's vocabulary and the verification record. We did not compile the earlier proof.

- *Brahmagupta: area is the shoelace value.* `Brahmagupta.area` is the absolute value of the shoelace expression. It is *not* identified with the Lebesgue measure of the convex hull of $\{A, B, C, D\}$; that bridge needs polygon-measure groundwork that Mathlib does not have. For a convex quadrilateral, which the headline theorems assume, the shoelace value is the geometric area, but this is not proved inside Lean. For the same reason, the classical statements of Layer 15, which use Lebesgue measure, are proved only under this identification as an explicit hypothesis (Section 9.1).

- *Brahmagupta: trigonometric Bretschneider only for convex quadrilaterals.* The angle form $K^2 = (s-a)(s-b)(s-c)(s-d) - abcd\cos^2\frac{\alpha+\gamma}{2}$ is proved in version 2, with the shoelace area, for convex quadrilaterals (diagonals crossing in their interiors), as `Brahmagupta.Classical.l15_bretschneider_shoelace` (Section 9.1); the version-1 files have only the angle-free diagonal form `Brahmagupta.sixteen_area_sq`, valid for every quadrilateral.

- *Brahmagupta: degenerate configurations.* The hypotheses of `Brahmagupta.brahmagupta` (`Cospherical` and two `Sbtw` conditions) do not force the four vertices to be distinct: for instance $A = B$ and $C = D$ on a diameter are allowed. In such collapsed cases the formula holds with both sides 0. The informal reading "convex cyclic quadrilateral" is accurate only when the vertices are also distinct (Section 5).

- *Brahmagupta: non-convex orderings.* Crossed (self-intersecting) orderings of four concyclic points are not covered. For them `Brahmagupta.brahmagupta_sq` does not apply, and

`Brahmagupta.area_sq_eq_iff_ptolemy` shows that the formula fails unless Ptolemy's equality holds for that ordering.

- *Miquel: model and degeneracies.* The theorem is stated in $\mathbb{C}$ with explicit coordinates and `dist`, not with Mathlib's `EuclideanGeometry.Sphere` or `EuclideanGeometry.Cospherical` API. Version 2 adds such restatements: `Miquel.l14_miquel_cospherical`, with `Collinear`, side lines and `Cospherical`, still in $\mathbb{C}$, and the classical statements in EuclideanSpace $\mathbb{R}$ (Fin 2) (`Miquel.Classical.cov_B2_b`; Section 9.1). Degenerate configurations (a collinear triangle, or D, E, F at a vertex) are excluded rather than handled. M is not proved distinct from D, E, F, and can coincide with one of them in tangent configurations. The Miquel-point corollaries (spiral similarity, the complete-quadrilateral and pentagon theorems) are not formalised.

- *Miquel: prior art.* Miquel's theorem is already formalised in Isabelle/HOL (AFP, August 2026) and in Coq's HighSchoolGeometry. Our development is not first; its distinguishing feature is that it needs no hypotheses beyond non-degeneracy of the triangle and of the side points.

- *Miquel: an unwitnessed lemma and a decoupled control.* `Miquel.rp_zero` has no hand-written witness; it is harness-certified satisfiable. `Miquel.miquel_negative_control` is a standalone numerical non-concyclicity fact with no formal link to `Miquel.miquel_example`; it supports the claim that membership of the third circle is a real constraint only by inspection.

- *Malfatti: not the conjecture.* We formalise the construction and the radius formula only. Malfatti's maximal-area conjecture is false (Goldberg 1967) and is not formalised, nor is the optimality of the greedy arrangement (Zalgaller and Los' 1994).

- *Malfatti: uniqueness.* Uniqueness of the Malfatti circles is not proved. Consequently the radius formula is proved for the constructed circles, and for an arbitrary triple satisfying the classical conditions only under uniqueness as a hypothesis (`Malfatti.Classical.cov_B4_b_of_unique`). Containment, listed here in version 1, is now proved: each closed disc lies in the closed triangle (`Malfatti.Classical.l15_malfatti_subset`, Section 9.1).

- *Malfatti: conditional bridges.* `Malfatti.K_bridge` holds only for $a, b, c \in (0, \pi/4)$, `Malfatti.D_bridge` only for $\theta \in (0, \pi/2)$, and the geometric bridges assume a 2-dimensional plane (Section 7).

- *Malfatti: four unused hypotheses.* Ablation finds four hypotheses that the proofs do not use: $0 < r$ in `Malfatti.radius_eq`, $p < 1$ in `Malfatti.dist_identity`, $q < 1$ in `Malfatti.mu_lt_one` and $i \neq j$ in `Malfatti.fin3_third` (for any $i, j \in$ Fin 3 some k avoids both). None affects soundness, but each lemma is stated more weakly than it could be, and the hypotheses should be dropped or justified before publication.

- *Malfatti: declarations without mutants.* The mutation harness generated no mutants for `Malfatti.isTangentAt_side`, `Malfatti.malfattiCircle_isExtTangent`, `Malfatti.malfattiCircle_tangent_side`, `Malfatti.sSameSide_center` and `Malfatti.touch_mem`, so the mutation layer says nothing about these five statements; they have been checked only by reading and by their use in the headline theorems.

- *Feuerbach and Brahmagupta: declarations without mutants.* The harness generated no mutants for the Feuerbach tangency theorems `Feuerbach.isIntTangent_`

`insphere_ninePointCircle` and `Feuerbach.isExtTangent_exsphere_ninePointCircle` or for `Feuerbach.nontrivial_of_triangle`, nor for Brahmagupta's headline statements `Brahmagupta.brahmagupta`, `Brahmagupta.brahmagupta_sq`, `Brahmagupta.area_sq_le`, `Brahmagupta.area_sq_eq_iff_ptolemy`, `Brahmagupta.brahmagupta_identity` and `Brahmagupta.not_cospherical_of_area_sq_ne`. The mutation layer says nothing about these nine statements; they have been checked by reading, by the witnesses and by the numerical scripts.

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